

Practical — Are You Rational?

Advanced Microeconomics 1 · Thursday 1 October, 11:00–13:00 · bring a laptop with R

What this session is

In the Session 1 quiz you divided 100 tokens between two accounts, one paying euros if a single coin flip lands heads and one if tails, twenty-five times at varying rates. Each decision was a point on a budget line: writing x_1 for the heads-claim and x_2 for the tails-claim, you chose a bundle with $p_1x_1 + p_2x_2 = 100$, at a different price vector each round.

In this session you will write, from a blank R script, the complete test of whether those 25 choices are consistent with maximizing *any* well-behaved utility function — and then quantify how far they fall short. No packages, no template code: everything is built from loops you write yourself, checked against a small example whose answer you are told. (The same test has been run on lab subjects, on a representative Dutch panel, and recently on GPT — Chen et al. 2023. Today it runs on you.)

Before you arrive (this is the whole preparation)

A laptop with a **working R installation** (RStudio or plain R, either is fine). Confirm it works by running two lines — any two lines. If R starts and `1 + 1` answers, you are ready. The class dataset `rounds.csv` is distributed at the start of the session; find your rows with your quiz ID.

The session, step by step

Part 1 — Look at the data (~15 min)

Load `rounds.csv` with `read.csv`; pull out your own 25 rows; verify $p_1x_1 + p_2x_2 = 100$ in every round (it will be — ask yourself why that makes it *untestable*); plot your 25 budget lines with your 25 chosen points on them.

Part 2 — Build the rationality test (~50 min)

Do not start on the real data. Start on three made-up rounds where the answer is known — the same numbers you will work *by hand* in Exercise Class 1:

round	p_1	p_2	x_1	x_2
1	2	1	3	2
2	1	2	2	3
3	1	1	4	1

The steps, each a few lines of loops:

1. **The cost table.** Build the $T \times T$ matrix E with $E[t, s] = p_1^t x_1^s + p_2^t x_2^s$: what bundle s would have cost at round t 's prices. Two nested **for** loops. The diagonal is what was actually spent.
2. **Revealed preference.** Two TRUE/FALSE matrices: $\mathbf{RD}[t, s]$ if $E[t, t] \geq E[t, s]$ (bundle s was affordable when t was chosen: t revealed at least as good as s) and $\mathbf{PD}[t, s]$ for the strict version. Add a tolerance of **1e-9** to every comparison.
3. **Chains.** Preference chains through intermediate rounds: extend $\mathbf{RD}$ by allowing each round k to act as a stepping stone — a third loop wrapped around the two you have (**if** $\mathbf{RC}[t, k]$ **and** $\mathbf{RC}[k, s]$ **then** $\mathbf{RC}[t, s] \leftarrow \mathbf{TRUE}$).
4. **Violations.** Count pairs where the chain says $t \succsim s$ while the data say s strictly beats t : $\mathbf{RC}[t, s] \ \&\& \ \mathbf{PD}[s, t]$.

Checkpoint: the toy data produce exactly 3 violations. Then overwrite the four input vectors with your own data ($T = 25$ now) and re-run the identical loops. The loops do not care.

Part 3 — The CCEI (~30 min)

Violation counts are crude. Afriat's question: *how much of your budget must be wasted before your choices become coherent?* Pretend the person had only a fraction e of what they spent: replace $E[t, t]$ by $e \cdot E[t, t]$ in step 2 — one new symbol. Then search: start at $e = 1$ and walk down in steps of 0.005, re-running your whole test, until the violations vanish. The first e that works is your **CCEI**; $1 - \text{CCEI}$ is the share of your budget your inconsistency burns.

Checkpoint: the toy data give CCEI = 0.875, exactly 7/8. Then: your own CCEI, and a loop over every ID in the file to build the class distribution — the histogram goes on the projector, with you marked on it.

Part 4 — Would this test catch anyone? (~15 min)

Before anyone brags: score 100 simulated *random* choosers — `runif(25, 0, 100)` tokens in H , converted to bundles exactly as the quiz was — with the identical code, on our actual prices (Bronars 1987). If random behavior scores well, a high CCEI means nothing. **Checkpoint: their mean CCEI is about 0.71 on our budgets** — so it doesn't, and yours does.

Deliverable

Two pages, due **[date — suggest: Thursday 8 October, before EC1]**, submitted via **[method]**. Contents: (i) your CCEI and where it sits in the class distribution; (ii) the random-chooser figure with one paragraph interpreting it; (iii) one disciplined paragraph: *what does a CCEI of 1 actually*

certify, and what evidence would make you distrust a high score? Consider the classmate who put all 100 tokens in the cheaper account every single round.

References

Varian (1982), *Econometrica*, §1–2; Bronars (1987), *Econometrica*; Afriat (1967); Choi, Fisman, Gale & Kariv (2007), *AER*; Choi, Kariv, Müller & Silverman (2014), *AER*; Chen, Liu, Shan, Zhong & Zhou (2023), *PNAS*; Chambers & Echenique, *Revealed Preference Theory*, Ch. 3–4.